

Urine Dipstick Analyzer v2.0 - Bill of Materials

ESP32-S3 | ILI9341 TFT | OV2640 Camera | DS3231 RTC | MAX17048 Battery Gauge

Ref	Component	MPN	Description	Qty	Unit \$	Total \$	Source		
U5	ESP32-S3-DevKitC-1-N8R8	ESP32-S3-DEVKITC-1-N8R8	Espressif ESP32-S3 DevKit 8MB Flash ..	1	\$15.00	\$15.00	DigiKey		Link
U10	ILI9341 2.4in TFT 320x24..	MSP2402	2.4in 320x240 SPI TFT LCD ILI9341 dr..	1	\$12.00	\$12.00	Amazon		Link
BT1	LiPo Battery 3.7V 1000mAh	LP503562	3.7V 1000mAh lithium polymer recharg..	1	\$10.00	\$10.00	Amazon		Link
U1	MCP73833 USB-C LiPo Char..	MCP73833-FCI/UN	Microchip MCP73833 Li-Ion/Li-Po char..	1	\$1.20	\$1.20	DigiKey		Link
U1a	USB-C Breakout Board	USB-C-BO-16P	USB Type-C 16-pin female breakout bo..	1	\$3.50	\$3.50	Amazon		Link
U2	Pololu 9V Step-Up/Down R..	S13V25F9	Pololu 9V 2.5A Step-Up/Step-Down Vol..	1	\$12.00	\$12.00	Pololu		Link
U3	Pololu 3.3V Step-Down Re..	S7V8F3	Pololu 3.3V 1A Step-Up/Step-Down Vol..	1	\$6.00	\$6.00	Pololu		Link
U4	MAX17048 LiPo Fuel Gauge	MAX17048G+T	Analog Devices MAX17048 1-Cell Fuel ..	1	\$2.80	\$2.80	DigiKey		Link
F1	500mA PTC Resettable Fuse	MF-MSMF050-2	Bourns 500mA 6V PTC resettable fuse ..	1	\$0.30	\$0.30	DigiKey		Link
PCB1	Custom Power Distributio..	N/A	2-layer FR4 PCB 50x30mm power hub wi..	1	\$5.00	\$5.00	OSH Park		Link
U6	SHT31-D Temp/Humidity Br..	ADA2857	Adafruit SHT31-DIS-B2.5KS breakout l..	1	\$12.00	\$12.00	Adafruit		Link
U7	BH1750 Light Sensor Module	GY-302	GY-302 BH1750FVI-TR digital I2C ambi..	1	\$4.50	\$4.50	Amazon		Link
U8	DS3231 RTC Module	ZS-042	ZS-042 DS3231SN TCXO RTC module AT24..	1	\$6.00	\$6.00	Amazon		Link
R_NTC	NTC Thermistor 10kohm B3..	NTCLE100E3103JB0	Vishay 10kohm ±5% NTC thermistor rad..	1	\$0.50	\$0.50	DigiKey		Link
SW3	MC-38 Magnetic Reed Switch	MC-38	MC-38 wired magnetic door sensor NO ..	1	\$4.00	\$4.00	Amazon		Link
SW2	Micro Limit Switch	D2F-01F	Omron D2F-01F subminiature micro swi..	1	\$2.50	\$2.50	DigiKey		Link
CAM1	OV2640 Camera Module	OV2640-V3	OV2640 2MP SXGA 24-pin DVP camera mo..	1	\$8.00	\$8.00	Amazon		Link
CAM1a	CPL Filter 16mm	N/A	16mm circular polarizer lens filter ..	1	\$5.00	\$5.00	Amazon		Link
U9	GM65 Barcode Scanner Mod..	GM65	Grow GM65 1D/2D barcode QR scanner T..	1	\$28.00	\$28.00	Amazon		Link
D1	White LED Array Module	WS2812B-4	4x WS2812B 5050 RGB LED PCB strip 5V..	1	\$3.00	\$3.00	Amazon		Link
D2	RGB LED 5mm Common Anode	OVLBR4C8	TT Electronics OVLBR4C8 5mm diffused..	1	\$0.80	\$0.80	DigiKey		Link
D3	UV-C LED 275nm	IN-C35DTDU4	Inolux IN-C35DTDU4 275nm UV-C LED 35..	1	\$15.00	\$15.00	DigiKey		Link
M1	NEMA 17 Stepper Motor 20mm	17HS08-1004S	StepperOnline NEMA17 20mm 1A 0.13Nm ..	1	\$12.00	\$12.00	Amazon		Link
U12	A4988 Stepper Driver Board	A4988	Allegro A4988SETTR-T carrier with he..	1	\$5.00	\$5.00	Amazon		Link
HTR1	Polyimide Heater Film 25..	PI-HTR-2550-12V	Kapton flexible heater film 25x50mm ..	1	\$8.00	\$8.00	Amazon		Link
FAN1	30mm 5V Brushless Fan	MC30060V2-000U-A99	Sunon 30x30x6mm 5VDC 0.35W vapo bear..	1	\$5.50	\$5.50	DigiKey		Link
BZ1	Active Piezo Buzzer 12mm	PS1240P02CT	TDK PS1240P02CT 5V active piezo buzz..	1	\$1.50	\$1.50	DigiKey		Link
Q1	N-MOSFET 2N7000 TO-92	2N7000	ON Semi 2N7000 60V 200mA N-ch MOSFET	1	\$0.45	\$0.45	DigiKey		Link
Q2	N-MOSFET IRLZ44N TO-220	IRLZ44NPBF	Infineon IRLZ44NPBF 55V 47A logic-le..	1	\$1.40	\$1.40	DigiKey		Link
Q3	N-MOSFET IRLZ44N TO-220	IRLZ44NPBF	Infineon IRLZ44NPBF 55V 47A logic-le..	1	\$1.40	\$1.40	DigiKey		Link
SW1	Tactile Switch 6x6mm	B3F-1000	Omron B3F-1000 tactile switch 6x6mm ..	1	\$0.30	\$0.30	DigiKey		Link
SD1	Micro SD Breakout Board ..	ADA254	Adafruit 254 Micro SD card breakout ..	1	\$7.50	\$7.50	Adafruit		Link
R1-R2	4.7kohm 0603 1% Resistor	RC0603FR-074K7L	Yageo 4.7kohm ±1% 0.1W 0603 I2C pull..	2	\$0.05	\$0.10	DigiKey		Link

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R3-R10	10kohm 0603 1% Resistor	RC0603FR-0710KL	Yageo 10kohm ±1% 0.1W 0603 pull-up/p..	6	\$0.05	\$0.30	DigiKey	Link
R7	47ohm 0805 1% Resistor	RC0805FR-0747RL	Yageo 47ohm ±1% 0.125W 0805 UV LED c..	1	\$0.05	\$0.05	DigiKey	Link
R11-R13	330ohm 0603 1% Resistor	RC0603FR-07330RL	Yageo 330ohm ±1% 0.1W 0603 RGB LED c..	3	\$0.05	\$0.15	DigiKey	Link
C1-C19	100nF 0603 X7R MLCC	GRM188R71C104KA01D	Murata 100nF 16V X7R 0603 decoupling..	19	\$0.10	\$1.90	DigiKey	Link
CB1-CB4	10uF 0805 X5R MLCC	GRM21BR61C106KE15K	Murata 10uF 16V X5R 0805 bulk bypass..	4	\$0.25	\$1.00	DigiKey	Link
CB5	100uF 1206 X5R MLCC	GRM31CR61A107ME15L	Murata 100uF 10V X5R 1206 motor supp..	1	\$1.50	\$1.50	DigiKey	Link
ENC1	Enclosure Base 180x120x4..	N/A	Custom injection-molded ABS base wit..	1	\$25.00	\$25.00	Protolabs	Link
ENC2	Enclosure Cover	N/A	Matching ABS cover with TFT window a..	1	\$7.00	\$7.00	Custom / 3D P	
TRAY1	Dipstick Tray 120x25x8mm	N/A	Custom POM tray with strip channel a..	1	\$10.00	\$10.00	Custom / 3D Print	
MNT1	Camera & LED Bracket	N/A	3D-printed ASA bracket for OV2640 + ..	1	\$4.00	\$4.00	3D Print FDM	
MNT2	Barcode Scanner Mount	N/A	3D-printed ASA bracket for GM65 alig..	1	\$3.00	\$3.00	3D Print FDM	
MNT3	Exhaust Vent Grill	N/A	3D-printed slotted grill for 30mm fa..	1	\$2.00	\$2.00	3D Print FDM	
MNT4	Cable Management Frame	N/A	Internal wire routing skeleton to ke..	1	\$2.00	\$2.00	3D Print FDM	
LS1	T8 Lead Screw 100mm + Nut	T8-100-CU	T8x2mm pitch 100mm lead screw + anti..	1	\$8.00	\$8.00	Amazon	Link
HW1	M2x6mm Pan Head Screws	ISO7045-M2x6-A2	M2x6 Phillips pan head A2 stainless ..	12	\$0.05	\$0.60	McMaster	Link
HW2	M3x8mm Pan Head Screws	ISO7045-M3x8-A2	M3x8 Phillips pan head A2 stainless ..	4	\$0.10	\$0.40	McMaster	Link
HW3	M3 Brass Heat-Set Inserts	94180A331	M3x5.7mm OD brass knurled heat-set i..	4	\$0.20	\$0.80	McMaster	Link
HW4	M2.5x11mm Brass Hex Stan..	N/A	M2.5 male-female brass hex standoff ..	8	\$0.50	\$4.00	Amazon	Link
GSK1	Silicone Gasket Sheet 1mm	N/A	Translucent silicone sheet 1mm 200x2..	1	\$3.00	\$3.00	Amazon	Link
GSK2	EPDM Foam Strip 3x10mm	N/A	EPDM closed-cell foam stripping 3x10..	1	\$5.00	\$5.00	Amazon	Link
OPT1	Frosted Acrylic Diffuser	N/A	White frosted acrylic 2mm sheet cut ..	1	\$2.00	\$2.00	Amazon	Link
OPT2	Fused Quartz Window 20x2..	N/A	Fused silica optical flat 20x20x1mm ..	1	\$8.00	\$8.00	Amazon	Link
OPT3	Telescope Flocking Paper	N/A	Self-adhesive black flocking paper A..	1	\$10.00	\$10.00	Amazon	Link
CAL1	X-Rite ColorChecker Nano	MSCCNANO	X-Rite ColorChecker Nano 24-patch mi..	1	\$25.00	\$25.00	Amazon	Link
CAL2	18% Gray / White Balance..	DGK-1	DGK Color Tools 18% gray + white bal..	1	\$5.00	\$5.00	Amazon	Link

TOTAL (111 parts)	\$329.95
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Electronics subtotal: \$0.00
Mechanical subtotal: \$329.95

Notes: Prices are approximate and based on single-unit purchases. Volume pricing (100+ units) typically reduces costs by 30-50%. The ILI9341 TFT, DS3231 RTC, and MAX17048 fuel gauge are v2.0 upgrades replacing the original SSD1306 OLED. Custom injection-molded parts (enclosure, tray) can be 3D-printed for prototyping.